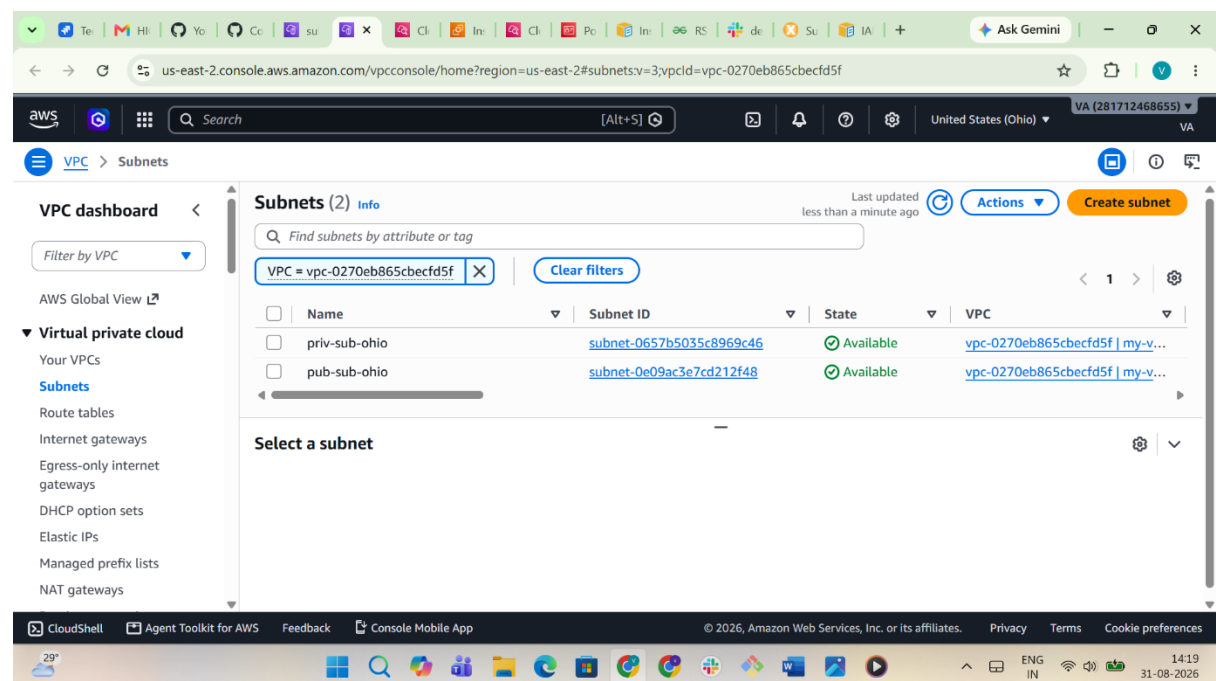


Task on VPC Transit Gateway & VPC Endpoint

1. Create one VPC, with 1 public subnet and 1 private subnet.

Implementation:

1. Created a VPC with a suitable CIDR block.
2. Created one public subnet and one private subnet within the VPC.
3. Created an Internet Gateway and attached it to the VPC.
4. Created a public route table and associated it with the public subnet.
5. Added a default route (0.0.0.0/0) to the Internet Gateway.
6. Created a private route table and associated it with the private subnet.
7. Verified the VPC, subnets, route tables, and Internet Gateway configuration.



2. Enable VPC peering for cross-region.

Implementation:

1. Created a VPC peering connection between VPCs located in different AWS Regions.
2. Specified the requester and acceptor VPCs with their respective Regions.
3. Accepted the VPC peering request from the destination Region.
4. Added routes to the respective route tables using the VPC peering connection.
5. Updated the security groups to allow communication between the VPC CIDR ranges.

6. Tested connectivity between resources in the cross-region VPCs.

The screenshot shows the AWS VPC console for the us-east-1 region. A green notification banner at the top states: "A VPC peering connection pcx-04c7bf112670d45cb / peering-virginia-ohio has been requested. Remember to change your region to us-east-2 to accept the peering connection." The main heading is "pcx-04c7bf112670d45cb / peering-virginia-ohio".

Details:

- Requester owner ID:** 281712468655
- Peering connection ID:** pcx-04c7bf112670d45cb
- Status:** Initiating Request to 281712468655
- Expiration time:** Monday, September 7, 2026 at 06:32:33 GMT+5:30
- Accepter owner ID:** 281712468655
- Requester VPC:** vpc-0795df3346611068d / my-vpc-01
- Requester CIDRs:** 192.168.0.0/24
- Requester Region:** N. Virginia (us-east-1)
- VPC Peering connection ARN:** arn:aws:ec2:us-east-1:281712468655:vpc-peering-connection/pcx-04c7bf112670d45cb
- Accepter VPC:** vpc-00c0a150e7f5dfff
- Accepter CIDRs:** -
- Accepter Region:** Ohio (us-east-2)

The screenshot shows the AWS VPC console for the us-east-2 region. A green notification banner at the top states: "Your VPC peering connection (pcx-04c7bf112670d45cb) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables." The main heading is "Peering connections (1)".

Peering connections (1):

Name	Peering connection ID	Status	Requester VPC
-	pcx-04c7bf112670d45cb	Provisioning	vpc-0795df3346611068d

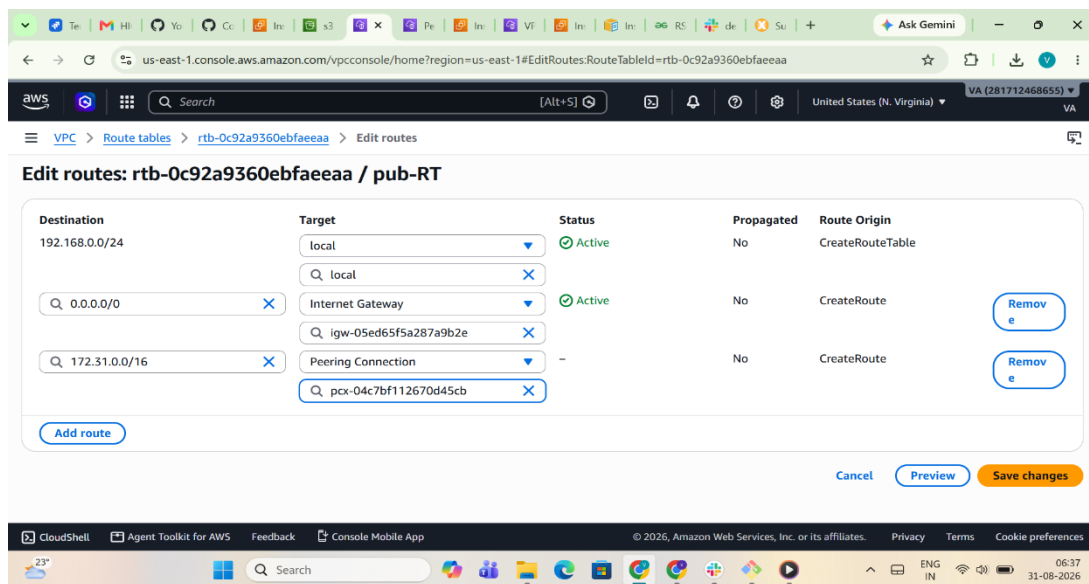
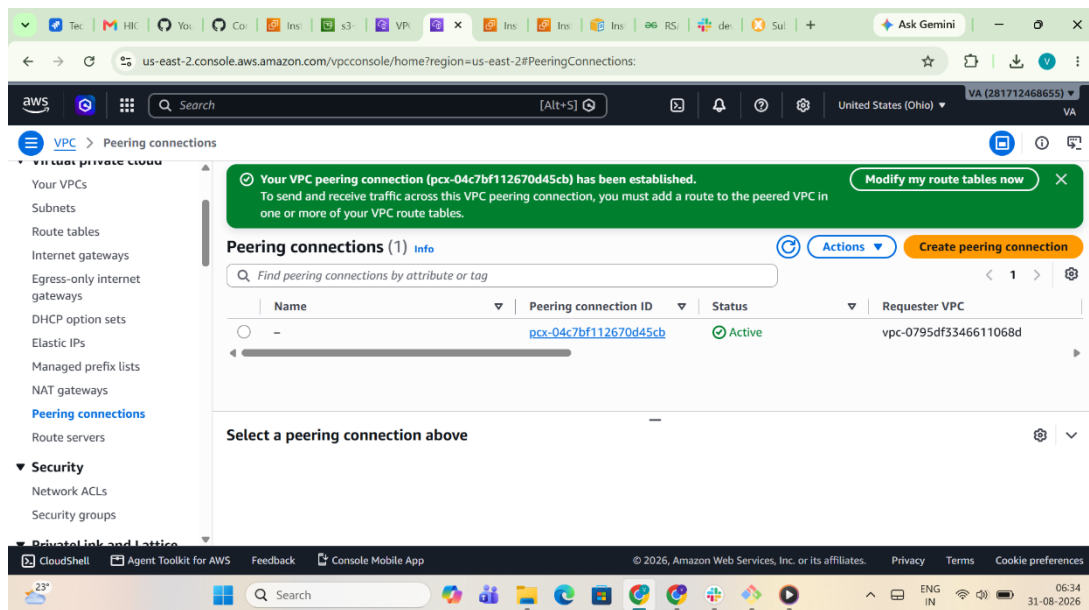
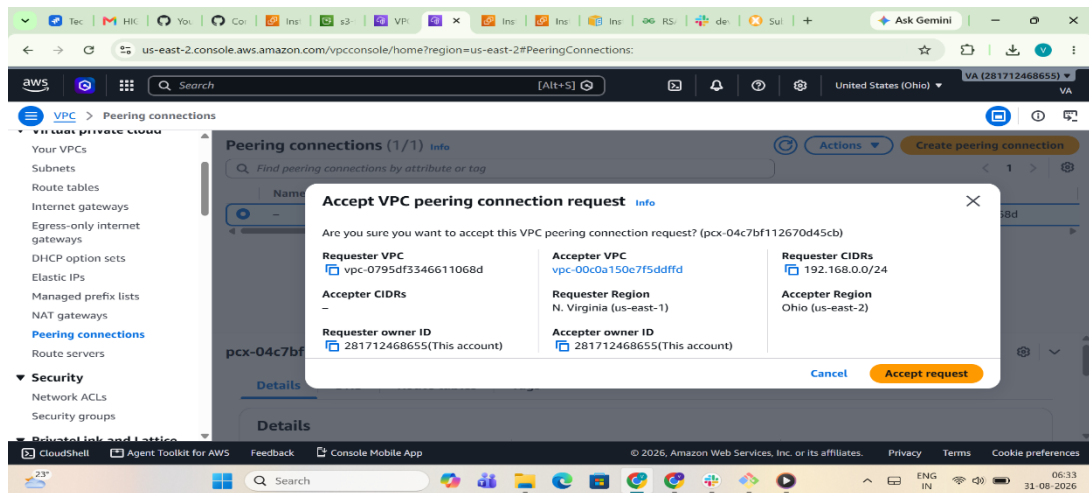
Buttons: [Modify my route tables now](#), [Create peering connection](#)

The screenshot shows the AWS VPC console for the us-east-2 region. A green notification banner at the top states: "Your VPC peering connection (pcx-04c7bf112670d45cb) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables." The main heading is "Peering connections (1)".

Peering connections (1):

Name	Peering connection ID	Status	Requester VPC
-	pcx-04c7bf112670d45cb	Pending acceptance	vpc-0795df3346611068d

Buttons: [Modify my route tables now](#), [Create peering connection](#)



The screenshot displays the AWS VPC console interface for editing routes in a specific route table. The console shows a table of routes with columns for Destination, Target, Status, Propagated, and Route Origin. The routes are configured for destinations 172.31.0.0/16, 0.0.0.0/0, and 192.168.0.0/24, all pointing to a Peering Connection target. Below the table, there are buttons for 'Add route', 'Cancel', 'Preview', and 'Save changes'.

Below the console screenshot, a terminal window shows the results of a ping command from an EC2 instance to a target IP address. The first ping attempt shows 100% packet loss, while the second attempt shows successful connectivity with 0% packet loss.

```
[ec2-user@ip-192-168-0-5 ~]$ ping 172.31.22.78
PING 172.31.22.78 (172.31.22.78) 56(84) bytes of data.
^C
--- 172.31.22.78 ping statistics ---
600 packets transmitted, 0 received, 100% packet loss, time 622926ms

[ec2-user@ip-192-168-0-5 ~]$ ping 172.31.22.78
PING 172.31.22.78 (172.31.22.78) 56(84) bytes of data.
64 bytes from 172.31.22.78: icmp_seq=1 ttl=127 time=10.6 ms
64 bytes from 172.31.22.78: icmp_seq=2 ttl=127 time=10.6 ms
64 bytes from 172.31.22.78: icmp_seq=3 ttl=127 time=10.6 ms
64 bytes from 172.31.22.78: icmp_seq=4 ttl=127 time=10.7 ms
64 bytes from 172.31.22.78: icmp_seq=5 ttl=127 time=10.7 ms
64 bytes from 172.31.22.78: icmp_seq=6 ttl=127 time=10.7 ms
64 bytes from 172.31.22.78: icmp_seq=7 ttl=127 time=10.7 ms
64 bytes from 172.31.22.78: icmp_seq=8 ttl=127 time=10.6 ms
64 bytes from 172.31.22.78: icmp_seq=9 ttl=127 time=10.6 ms
64 bytes from 172.31.22.78: icmp_seq=10 ttl=127 time=10.7 ms
^C
--- 172.31.22.78 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9016ms
rtt min/avg/max/mdev = 10.643/10.661/10.725/0.023 ms
[ec2-user@ip-192-168-0-5 ~]$
```

3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Implementation:

1. Created a VPC peering request from the VPC in the first AWS account.
2. Specified the VPC ID and AWS account ID of the peer account.
3. Accepted the VPC peering request from the second AWS account.
4. Added routes to both VPC route tables using the VPC peering connection.
5. Updated the security groups to allow traffic between the two VPCs.
6. Verified connectivity between resources in the cross-account VPCs.

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#PeeringConnectionDetails.vpcPeeringConnectionId=pcx-07175b393996abc66

A VPC peering connection pcx-07175b393996abc66 / peering-from-other has been requested.
The owner of vpc-0f3216a7937311169 must accept the peering connection.

pcx-07175b393996abc66 / peering-from-other

Details

Requester owner ID 281712468655	Accepter owner ID 956035085469	VPC Peering connection ARN arn:aws:ec2:us-east-1:281712468655:vpc-peering-connection/pcx-07175b393996abc66
Peering connection ID pcx-07175b393996abc66	Requester VPC vpc-0795df3346611068d / my-vpc-01	Accepter VPC vpc-0f3216a7937311169
Status Pending Acceptance by 956035085469	Requester CIDRs 192.168.0.0/24	Accepter CIDRs -
Expiration time Monday, September 7, 2026 at 11:57:02 GMT+5:30	Requester Region N. Virginia (us-east-1)	Accepter Region N. Virginia (us-east-1)

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#PeeringConnections:

Peering connections (1)

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC
peering-from-other	pcx-07175b393996abc66	Active	vpc-0795df3346611068d / my...

Select a peering connection above

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#PeeringConnections:

Your VPC peering connection (pcx-07175b393996abc66) has been established.
To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.

Peering connections (1)

Find peering connections by attribute or tag

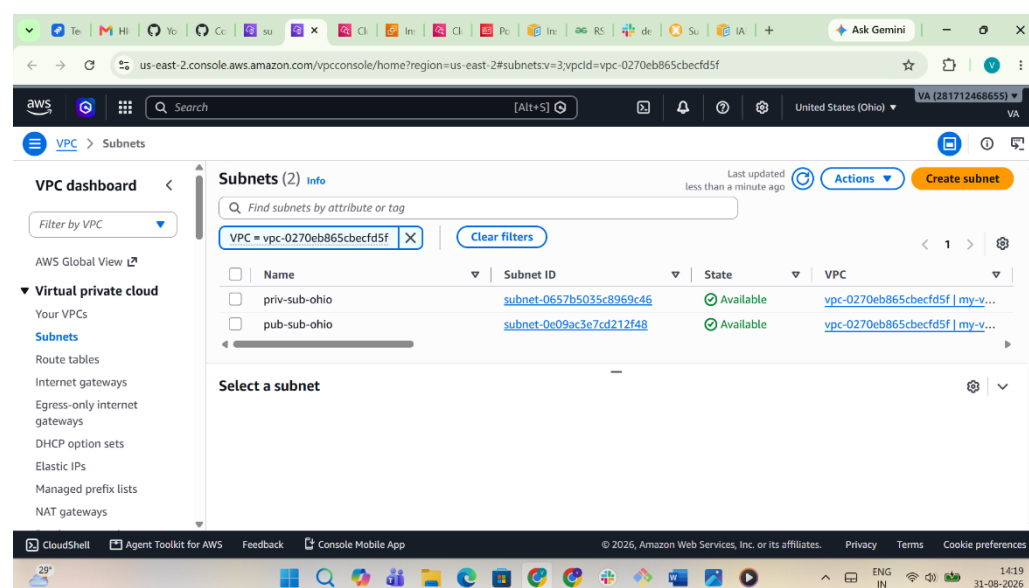
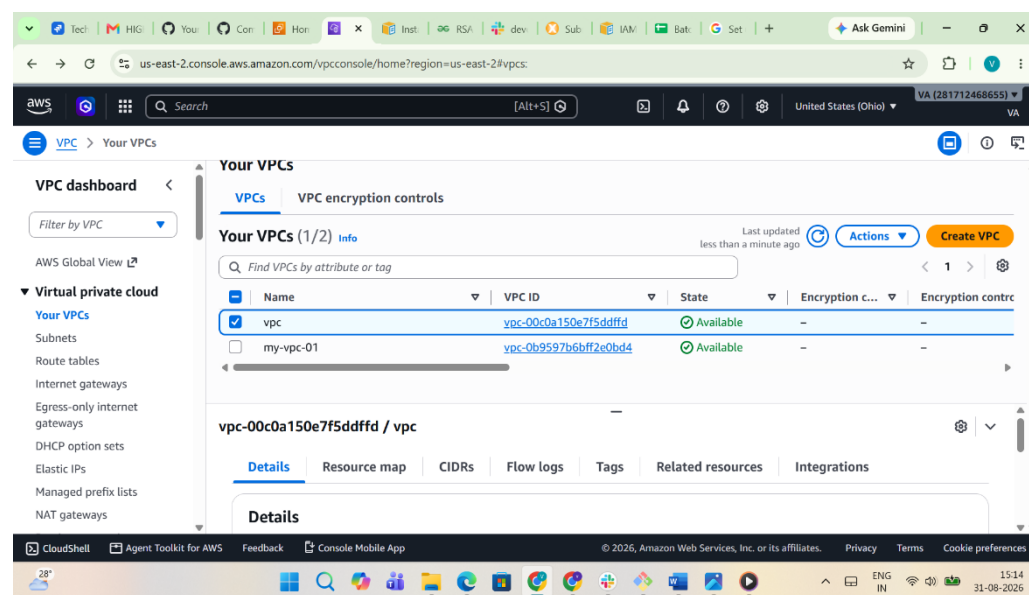
Name	Peering connection ID	Status	Requester VPC	Accepter VPC
-	pcx-07175b393996abc66	Active	vpc-0795df3346611068d	vpc-0f3216a7937311169

Select a peering connection above

4. Set up a VPC Transit Gateway.

Implementation:

1. Created a Transit Gateway in the AWS Region.
2. Created a Transit Gateway attachment for the VPC.
3. Selected the required subnets for the Transit Gateway attachment.
4. Associated the VPC attachment with the Transit Gateway route table.
5. Added routes in the VPC route tables pointing to the Transit Gateway.
6. Verified the Transit Gateway attachment and routing configuration.



us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTables:

Route tables (1/2) Info

Find route tables by attribute or tag

Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
rtb-07067717bd80690e8	subnet-0d980ca8ccdd6a...	-	Yes	vpc-00c0a150e7f5dddfd vpc
rtb-06515ce0d726183d7	subnet-0d1af0b37d7171...	-	Yes	vpc-0b9597b6b6f2e0bd4 my-v...

rtb-06515ce0d726183d7

Details Routes Subnet associations Edge associations Route propagation Tags Related

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#Instances:v=3,instanceState=running

Instances (2) Info

Filter instance by attribute or tag (case-sensitive)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
ec2-server2	i-037ff4830ceb28bce	Running	t3.micro	3/3 checks passed	us-east-2a
ec2-server1	i-01b857db13861c9f0	Running	t3.micro	3/3 checks passed	us-east-2b

All instances

Instance alarm states

Service health

Health events affecting EC2 instances

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#TransitGateways:

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

TGW-Lab

Transit gateways (1) Info

Find transit gateway by attribute or tag

Name	Transit gateway ID	Owner ID	State
TGW-Lab	tgw-0fd7d0bdedf37005d	281712468655	Pending

Select a transit gateway

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#TransitGateways:

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

TGW-Lab

Transit gateways (1) Info

Find transit gateway by attribute or tag

Name	Transit gateway ID	Owner ID	State
TGW-Lab	tgw-0fd7d0bdedf37005d	281712468655	Available

Select a transit gateway

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#TransitGatewayAttachments:v=3;\$case=tags:false;\$C.client:false;\$regex=tag...

Transit gateway attachments (2)

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource
TGW-VPCC2-Attachm...	tgw-attach-043dae672ba239687	tgw-0fd7d0bdebf37005d	Available	VPC
TGW-VPCC1-Attachm...	tgw-attach-06465049bda16f4f8	tgw-0fd7d0bdebf37005d	Available	VPC

Select a transit gateway attachment

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-06515ce0d726183d7

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Updated routes for rtb-06515ce0d726183d7 successfully

Details

Route	Yes	Subnet
rtb-06515ce0d726183d7	Yes	subnet-0d1af0b37d7171bd3 / my-subnet

VPC: [vpc-0b9597b6bffa2e0bd4](#) | my-vpc-01

Owner ID: [281712468655](#)

Routes (3)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0312b8e8d81eb3...	Active	No	Create Route
172.31.16.0/20	tgw-0fd7d0bdebf370...	Active	No	Create Route
172.168.0.0/24	local	Active	No	Create Route Table

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-07067717bd80690e8

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Updated routes for rtb-07067717bd80690e8 successfully

Details

Route	Yes	Subnet
rtb-07067717bd80690e8	Yes	subnet-0d50c480c00000002

VPC: [vpc-00c0a150e7f5d0ff4](#) | vpc

Owner ID: [281712468655](#)

Routes (3)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-058f0884d5f0c539f	Active	No	Create Route
172.31.0.0/16	local	Active	No	Create Route Table
172.168.0.0/28	tgw-0fd7d0bdebf370...	Active	No	Create Route

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0a772fd9d0dad4114

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Details

Route table ID: rtb-0a772fd9d0dad4114

VPC: vpc-0b9597b6bf2e0bd4 | my-vpc-01

Owner ID: 281712468655

Main: No

Explicit subnet associations: -

Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
pl-7ba54012	vpc-06a9acef507ad3...	Active	No	Create Route
172.168.0.0/24	local	Active	No	Create Route Table

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-037ff4830ceb28bce

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instance summary for i-037ff4830ceb28bce (ec2-server2)

Updated less than a minute ago

Instance ID: i-037ff4830ceb28bce

IPV6 address: -

Hostname type: IP name: ip-172-168-0-14.us-east-2.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: 18.226.87.152 [Public IP]

Public IPv4 address: 18.226.87.152 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-168-0-14.us-east-2.compute.internal

Instance type: t3.micro

VPC ID: vpc-0b9597b6bf2e0bd4 (my-vpc-01)

Private IPv4 addresses: 172.168.0.14

Public DNS: -

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-09a5b2cc4d3ef8296

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instance summary for i-09a5b2cc4d3ef8296 (my-priv-ec2)

Updated less than a minute ago

Instance ID: i-09a5b2cc4d3ef8296

IPV6 address: -

Hostname type: IP name: ip-172-168-0-24.us-east-2.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: 16.59.54.82 [Public IP]

Public IPv4 address: 16.59.54.82 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-168-0-24.us-east-2.compute.internal

Instance type: t3.micro

VPC ID: vpc-0b9597b6bf2e0bd4 (my-vpc-01)

Private IPv4 addresses: 172.168.0.24

Public DNS: -

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/roles/details/EC2-S3-Role

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Last activity: 10 minutes ago

Maximum session duration: 1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (2)

You can attach up to 20 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
AmazonS3ReadOnlyAccess	AWS managed	1
ec2-s3-policy	Customer inline	0

```

~\_#- Amazon Linux 2023
~~\#####
~~\#####\
~~\###|
~~\#/
~~\_~'~>

```

```
[ec2-user@ip-172-168-0-24 ~]$ aws sts get-caller-identity
{
  "UserId": "AROAUDF27EKXVBATC4DN:i-09a5b2cc4d3ef8296",
  "Account": "281712468655",
  "Arn": "arn:aws:sts:281712468655:assumed-role/EC2-S3-Role/i-09a5b2cc4d3ef8296"
```

← → ↺ 🌐 us-east-2.console.aws.amazon.com/s3/buckets/my-vpc-endpoint-test-281712468655?region=us-east-2&tab=objects

✔ Successfully created bucket "my-vpc-endpoint-test-281712468655"

my-vpc-endpoint-test-281712468655 [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	 test.txt	txt	August 31, 2026, 17:27:45 (UTC+05:30)	24.0 B	Standard